Miles Smith

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Research Interests

My research sits at the intersection of biomechanics, neuromechanics, and motor control, with a focus on understanding how the nervous system structures movement to accomplish complex tasks. I combine computational modeling with experiments to study neural control strategies in locomotion, with applications to clinical assessment and health monitoring.

Education

2023 – Present **Massachusetts Institute of Technology** – Cambridge, Massachusetts
SM/PhD in Mechanical Engineering
Advisor: Brian Anthony
SM Thesis: *Computational Neuromechanics of Impaired Locomotion: Using Predictive Neuromusculoskeletal Simulations to Unravel the Structure of Variability*

2020 – 2022 **Stanford University** – Stanford, California
MS in Civil and Environmental Engineering (Atmosphere/Energy)

2016 – 2020 **University of Maryland, Baltimore County** – Baltimore, Maryland
BS in Mechanical Engineering

Work Experience

09/2023 – Present **Graduate research assistant at MIT**
Advisor: Dr. Brian Anthony

2022 – 2023 **Mechanical engineer at Otherlab/Copper Home**
Supervisor: Dr. Samuel Calisch

2021 – 2022 **GEM research fellow at MIT Lincoln Laboratory**
Supervisor: Dr. Theodore Bloomstein

Honors and scholarships

2024 Wunsch Foundation Silent Hoist and Crane Award (MIT)

2021 USTFCCCA NCAA Division I Men's Track & Field All-Academic Award (Stanford)

2020 GEM Full Fellowship (Stanford)

2020 Arthur Ashe Jr. Sports Scholar Award (UMBC)

2016 Meyerhoff Premier Scholarship Award (UMBC)

Selected Publications

For a full list of publications, please see Google Scholar.

2026 **Neuromuscular impairments alter energetic cost landscape curvature and stride speed variability in post-stroke locomotion**
Smith, Miles; Namburi, Praneeth; Seethapathi, Nidhi; Anthony, Brian
PLOS Computational Biology (Submitted for publication.)

2026 **Revealing solid electrolyte interphase at the Li/electrolyte interface in Li-ion Batteries by in situ Fourier Transform Infrared Spectroscopy**
Wang, Daniel; Katayam, Yu; Von Holtum, Bastian; Svirinovsky-Arbeli, Asya; **Smith, Miles**; … Shao-Horn, Yang
Cell Press Blue

Patents

2025 Battery-integrated appliance system and method (US20250071863A1)

2025 Battery-integrated water heater system and method (US20250075943A1)

2024 Induction Heating Adapter System and Method (US20240206023A1)

Technical Skills

Programming languages: MATLAB, Python, C, C++, JavaScript, HTML/CSS

Software: LaTeX, Git, OpenSim, MuJoCo

Hardware: ESP-32, Arduino

Teaching, Mentorship, and Service

2025 – Present Founding vice president of Melanated MechEs (MIT)

Fall 2025 Lab instructor, 2.00: Introduction to Design (MIT)

Conferences and Workshops

March 2026 A Robust, High-Throughput Framework for Predictive Simulations Using *MocoStudy*
OpenSim+ Advanced Workshop